# Multi-Factor Attrition Analysis and Employee Engagement Profiling

*Corporate HR Analytics & Workforce Attrition*

School of Computer Science | IBM HR Analytics Dataset

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## I. Abstract

Employee attrition is a critical challenge faced by modern organizations, resulting in significant operational, financial, and knowledge-related costs. This study presents a comprehensive multi-factor analysis of workforce attrition using the IBM HR Analytics Employee Attrition and Performance Dataset. The research investigates the key determinants of attrition— including monthly income, overtime practices, work-life balance, job satisfaction, departmental affiliation, and total working experience— through systematic Exploratory Data Analysis (EDA), statistical hypothesis testing, and predictive modelling. Results indicate that overtime, low salary, poor work-life balance, and low job satisfaction are the most significant predictors of employee departure. Employees in Sales and Human Resources departments exhibit the highest attrition vulnerability. The study concludes with actionable retention strategies and a risk-based employee segmentation model categorized into Low, Medium, and High attrition risk groups. These findings provide HR professionals and business analysts with a data-driven framework for proactive workforce management.

## II. Introduction

Workforce attrition represents one of the most pressing concerns in contemporary human resource management. When experienced employees leave an organization, the direct and indirect costs— including recruitment expenses, onboarding time, lost institutional knowledge, and reduced team productivity— can amount to 50–200% of an employee's annual salary. Despite these consequences, many organizations still rely on reactive rather than proactive retention strategies.

The advent of people analytics and machine learning offers unprecedented opportunities to transform HR decision-making. By systematically analysing historical workforce data, organizations can identify at-risk employees before they resign, understand the root causes of dissatisfaction, and allocate retention resources more effectively.

This project leverages the publicly available IBM HR Analytics Dataset to conduct a structured, multi-layered investigation into attrition patterns. The research objectives are:

- To identify which employee demographics and workplace factors most strongly predict attrition.
- To validate statistical relationships between key variables using chi-square tests and ANOVA.
- To develop an employee risk-profiling model with recommended retention actions.
- To provide actionable business intelligence for HR stakeholders.

## III. Literature Review

Several studies have investigated the determinants of employee attrition using quantitative and qualitative methodologies. Mobley (1977) proposed one of the earliest models of employee turnover, emphasizing job dissatisfaction as the primary antecedent of resignation behaviour. Subsequent research by Price (2001) expanded this framework to include organizational commitment, pay equity, and work-family conflict as independent predictors.

In the domain of machine learning, Bhatt et al. (2021) demonstrated that Random Forest classifiers achieve over 87% accuracy in predicting attrition when trained on HR datasets with features including overtime, job level, and work-life balance. Similarly, Fallucchi et al. (2020) applied Logistic Regression and Decision Trees to IBM HR data, finding that monthly income and years at the company were among the top five predictive features.

Studies focusing on overtime specifically (Pencavel, 2015) found a strong nonlinear relationship between working hours and productivity, suggesting that excessive overtime correlates with both health decline and increased resignation intent. These findings align with the exploratory patterns observed in the present dataset.

Work-life balance has been increasingly highlighted in post-pandemic HR research (Microsoft Work Trend Index, 2022), with 47% of employees citing poor work-life balance as a primary reason for seeking new employment. The current study operationalizes this variable as a numeric satisfaction rating from 1 (low) to 4 (high) and examines its distributional relationship with attrition outcomes.

## IV. Dataset Description

### A. Source

The IBM HR Analytics Employee Attrition & Performance Dataset is a synthetic, publicly available dataset created by IBM data scientists and hosted on Kaggle. It is designed to reflect realistic HR scenarios.

| Attribute | Details |
|---|---|
| Total Records | 1,470 employees |
| Total Features | 35 columns |
| Target Variable | Attrition (Yes / No) |
| Attrition Rate | ~16.12% (237 Yes, 1,233 No) |
| Data Type | Mixed (numerical + categorical) |
| Missing Values | None |
| Duplicates | None detected |

### B. Key Features

The following table summarises the primary features used in analysis:

| Feature | Description |
|---|---|
| Age | Employee age (18–60 years) |
| MonthlyIncome | Monthly salary in USD |
| OverTime | Whether employee works overtime (Yes/No) |
| JobSatisfaction | Rating 1–4 (Low to Very High) |
| WorkLifeBalance | Rating 1–4 (Bad to Best) |
| Department | HR, Sales, Research & Development |
| TotalWorkingYears | Total career experience in years |
| YearsAtCompany | Tenure at IBM |
| DistanceFromHome | Commute distance in km |
| EnvironmentSatisfaction | Workplace environment rating 1–4 |
| BusinessTravel | Non-Travel, Travel_Rarely, Travel_Frequently |

## V. Data Cleaning

### A. Missing Values

A thorough audit of the dataset using df.isnull().sum() confirmed zero missing values across all 35 columns. No imputation was required. This is consistent with the dataset's synthetic origin, which was designed to be analysis-ready.

### B. Duplicate Detection

No duplicate employee records were identified using df.duplicated().sum(). The dataset integrity was preserved throughout the analysis.

### C. Constant & Irrelevant Columns

Three columns were identified as having zero variance (constant values across all rows): EmployeeCount (always 1), StandardHours (always 80), and Over18 (always Y). These features provide no discriminatory information for modelling and were dropped prior to analysis.

### D. Outlier Analysis

The Interquartile Range (IQR) method was applied to the three numerically skewed variables: MonthlyIncome, TotalWorkingYears, and YearsAtCompany. The IQR formula defines outliers as values falling below Q1 - 1.5×IQR or above Q3 + 1.5×IQR.

MonthlyIncome exhibited upper outliers corresponding to senior executives and directors, which is expected in realistic salary distributions. These values were retained as they reflect genuine compensation structures rather than data errors.

### E. Log Transformation

Due to the right-skewed nature of MonthlyIncome (skewness > 1.5), a log transformation was applied using df['Income_Log'] = np.log1p(df['MonthlyIncome']). This normalized the distribution, reduced the influence of executive salary outliers, and improved the performance of downstream statistical and ML models that assume normality or linearity.

## VI. Exploratory Data Analysis (EDA)

### A. Univariate Analysis

Age Distribution: The workforce age distribution follows a near-normal pattern, concentrated in the 30–40 year range. The median age is approximately 36 years, indicating a predominantly mid-career workforce.

Monthly Income: The income distribution is heavily right-skewed, with most employees earning between $2,000–$6,000/month. A small proportion of senior roles earn $15,000–$20,000, forming a visible upper tail. Post-log-transformation, the distribution approximates normality.

Attrition Distribution: The dataset is class-imbalanced. Approximately 83.88% of employees (1,233) did not leave, while 16.12% (237) left the organization. This imbalance was addressed using stratified sampling and SMOTE during model training.

Departmental Distribution: Research & Development constitutes the largest department (~65%), followed by Sales (~30%) and Human Resources (~5%).

### B. Bivariate Analysis

Attrition vs. Monthly Income: Box plots reveal that employees who left the organization had a significantly lower median monthly income (~$4,787) compared to those who stayed (~$6,833). This strongly suggests that compensation inadequacy is a key attrition driver.

Attrition vs. Overtime: Cross-tabulation analysis shows that 53.7% of employees who worked overtime eventually left, compared to only 10.4% of non-overtime workers. This is the strongest single categorical predictor of attrition in the dataset.

Attrition vs. Work-Life Balance: Employees rating their work-life balance as 1 (Bad) show an attrition rate of ~31%, compared to ~14% for those rating it 3 (Good). Poor work-life balance is a statistically and practically significant attrition predictor.

Attrition vs. Job Satisfaction: Employees with Job Satisfaction of 1 (Low) exhibit the highest attrition (~22.8%), while those with satisfaction of 4 (Very High) show the lowest (~11.3%).

### C. Correlation Analysis

A Pearson correlation heatmap was generated for all numeric variables. Key observations:

- MonthlyIncome is strongly positively correlated with TotalWorkingYears (r = 0.77) and JobLevel (r = 0.95), confirming that experience and seniority drive compensation.
- YearsAtCompany and YearsWithCurrManager show moderate positive correlation (r = 0.77), indicating that long-tenured employees tend to have stable manager relationships.
- Attrition (binary encoded) shows a negative correlation with MonthlyIncome (-0.16) and JobLevel (-0.17), and a positive correlation with OverTime (encoded: 0.25).

## VII. Statistical Validation

### A. Chi-Square Test— Overtime vs. Attrition

Null Hypothesis ($H_0$): There is no significant association between overtime and attrition.

Alternative Hypothesis ($H_1$): Overtime is significantly associated with attrition.

| Statistic | Value | Interpretation |
|---|---|---|
| Chi-Square ($\chi^2$) | 87.61 | High test statistic |
| p-value | < 0.0001 | **Reject $H_0$ — Significant** |
| Degrees of Freedom | 1 | 2×2 contingency table |

Conclusion: There is a statistically significant association between overtime and attrition ($p < 0.05$). $H_0$ is rejected. Overtime is a major attrition driver.

### B. ANOVA— Salary Across Departments

Null Hypothesis ($H_0$): Mean monthly income is equal across all three departments (HR, Sales, R&D).

Alternative Hypothesis ($H_1$): At least one department has a significantly different mean salary.

| Department | Mean Monthly Income | Attrition Rate |
|---|---|---|
| Human Resources | $6,655 | **~25.9%** |

| | | |
|---|---|---|
| Sales | $6,959 | **~20.6%** |
| Research & Development | $6,281 | ~13.8% |

The ANOVA F-statistic is significant ($p < 0.05$), indicating that salary differs meaningfully across departments. R&D, despite having the lowest mean salary, has the lowest attrition— suggesting that intrinsic motivation, job nature, and career growth opportunities compensate for lower pay in that department.

# VIII. Results

## A. Key Findings

| Finding | Insight |
|---|---|
| Highest Attrition Dept. | Human Resources (~25.9% attrition rate) |
| Overtime Impact | 53.7% of overtime employees left vs. 10.4% non-OT |
| Salary Threshold | Employees earning < $3,500/month show 3× higher attrition |
| Work-Life Balance | Rating 1 (Bad) leads to ~31% attrition vs. ~14% at Rating 3 |
| High-Risk Job Roles | Sales Representatives, Lab Technicians, HR Executives |
| Age Group Most at Risk | Employees aged 18–30 (early career, high mobility) |
| Tenure Sweet Spot | Employees with <2 years or 5–8 years tenure most likely to leave |

## B. Employee Risk Segmentation

Based on the multivariate analysis, employees were segmented into three risk tiers:

| Risk Level | Profile Characteristics | Recommended Action |
|---|---|---|
| **High Risk** | OT=Yes, Income<$3,500, WLB=1, Satisfaction=1 | Immediate salary review, OT reduction, 1-on-1 HR |
| **Medium Risk** | Moderate salary, occasional OT, WLB=2, Satisfaction=2 | Flexible work, training programs, recognition |
| **Low Risk** | OT=No, High income, WLB=3-4, Satisfaction=3-4 | Career growth planning, mentorship roles |

# IX. Recommendations

## A. Compensation Strategy

- Establish a salary floor of at least $4,000/month for entry and junior-level employees to reduce the primary economic attrition trigger.

- Conduct market benchmarking annually and adjust compensation for Sales and HR roles, which exhibit the highest attrition despite mid-range salaries.
- Introduce performance-linked bonuses in lieu of mandatory overtime compensation to align employee incentives with business outcomes.

### B. Overtime Policy

- Implement a maximum overtime cap of 5 hours per week for consecutive weeks and flag employees who exceed this threshold for HR intervention.
- Introduce compensatory off days and remote working options for employees required to work beyond standard hours.
- Develop an overtime monitoring dashboard for managers to redistribute workloads proactively.

### C. Work-Life Balance Initiatives

- Introduce flexible working hours (flexitime), hybrid work arrangements, and mandatory annual leave consumption policies.
- Conduct quarterly pulse surveys to track work-life balance satisfaction and take corrective action when department scores fall below a threshold of 2.5.

### D. Departmental Interventions

- Human Resources department requires immediate structural intervention: role enrichment, career path clarity, and competitive salary benchmarking.
- Sales department should receive dedicated employee wellness programs, given that commission-based income structures increase financial stress and attrition risk.

### E. Predictive Retention System

- Deploy a Random Forest or XGBoost-based attrition prediction model in the HRMS system to flag at-risk employees on a monthly basis.
- Assign dedicated HR business partners to monitor and engage high-risk employee segments with personalised retention plans.

## X. Future Enhancements

- Real-time data integration with HR systems
- Use advanced ML models (XGBoost, Deep Learning)
- NLP sentiment analysis for employee feedback
- Interactive dashboards (Power BI / Tableau)
- Time-series forecasting for future attrition
- Deploy as web application with alert system

## XI. Conclusion

This research presents a comprehensive, data-driven analysis of employee attrition using the IBM HR Analytics dataset. Through systematic EDA, statistical hypothesis testing, and risk-based segmentation, the study established that overtime, low compensation, poor work-life balance, and low job satisfaction are the primary drivers of employee departure. Human Resources and Sales departments were identified as the most vulnerable units requiring immediate retention intervention.

The Chi-Square test confirmed a highly significant association between overtime and attrition ($p < 0.0001$), while ANOVA established meaningful salary variation across departments. Employee risk profiling categorised the workforce into three segments— High, Medium, and Low Risk— with tailored retention strategies for each.

The practical recommendations derived from this study— including salary benchmarking, overtime policy reform, flexible work arrangements, and predictive HR systems— provide a deployable framework for organizations seeking to reduce attrition rates and improve workforce stability. Future work will extend this analysis to incorporate NLP-based sentiment analysis of employee exit interviews and longitudinal attrition forecasting using time-series modelling.

## XII. References


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